2D Arrays Master Practice Sheet

2D Arrays Practice Sheet

This sheet includes a set of carefully selected 2D array problems categorized by difficulty. Each problem includes a clear statement, sample input/output, and difficulty level. Use these for strong logic-building practice before your quiz.

1. Sum of All Elements

Difficulty: Easy

Write a program to input a 2D array and print the sum of all its elements.

Sample Input:

3 3

1 2 3

4 5 6

7 8 9

Sample Output:

Sum = 45

2. Row-wise Sum

Difficulty: Easy

Given a 2D array, print the sum of each row on a separate line.

Sample Input:

2 3

1 2 3

4 5 6

Sample Output:

Row 1 Sum = 6

Row 2 Sum = 15

3. Transpose of Matrix

Difficulty: Easy

Write a program to transpose a 2D matrix (swap rows with columns).

Sample Input:

2 3

1 2 3

4 5 6

Sample Output:

1 4

2 5

3 6

4. Search an Element

Difficulty: Easy

Search a given number in a 2D array and print its index (row, col) if found.

Sample Input:

3 3

1 2 3

4 5 6

7 8 9

Search: 6

Sample Output:

Found at position: (1, 2)

5. Matrix Addition

Difficulty: Medium

Take two matrices of the same size and print their sum matrix.

Sample Input:

2 2

Matrix A:

1 2

3 4

Matrix B:

5 6

7 8

Sample Output:

6 8

10 12

6. Matrix Multiplication

Difficulty: Medium

Take two matrices and multiply them (only if col1 == row2).

Sample Input:

Matrix A (2x3):

1 2 3

4 5 6

Matrix B (3x2):

7 8

9 10

11 12

Sample Output:

58 64

139 154

7. Identity Matrix Checker

Difficulty: Medium

Check if a given square matrix is an identity matrix.

Sample Input:

3 3

1 0 0

0 1 0

0 0 1

Sample Output:

Yes, it is an identity matrix.

8. Diagonal Sum

Difficulty: Medium

Print the sum of the main and secondary diagonals of a square matrix.

Sample Input:

3 3

1 2 3

4 5 6

7 8 9

Sample Output:

Main Diagonal = 15

Secondary Diagonal = 15

9. Spiral Print

Difficulty: Hard

Print elements of a 2D matrix in spiral order (clockwise from top left).

Sample Input:

3 3

1 2 3

4 5 6

7 8 9

Sample Output:

1 2 3 6 9 8 7 4 5

10. Rotate Matrix 90 Degrees

Difficulty: Hard

Rotate a square matrix 90 degrees clockwise.

Sample Input:

3 3

1 2 3

4 5 6

7 8 9

Sample Output:

7 4 1

8 5 2

9 6 3

Extended Practice Questions

11. Count Even and Odd Numbers

Difficulty: Easy

Count how many even and odd numbers are present in a given 2D matrix.

Sample Input:

2 3

1 2 3

4 5 6

Sample Output:

Even: 3, Odd: 3

12. Maximum Element in Each Row

Difficulty: Easy

Find and print the maximum element from each row.

Sample Input:

2 3

4 9 2

1 7 6

Sample Output:

Max of Row 1: 9

Max of Row 2: 7

13. Saddle Point in a Matrix

Difficulty: Medium

Find the saddle point of the matrix (min in row and max in its column).

Sample Input:

3 3

1 2 3

4 5 6

7 8 9

Sample Output:

Saddle Point: 7

14. Boundary Traversal

Difficulty: Medium

Print only the boundary elements of a matrix in a clockwise manner.

Sample Input:

3 3

1 2 3

4 5 6

7 8 9

Sample Output:

1 2 3 6 9 8 7 4

15. Print Zig-Zag Pattern

Difficulty: Medium

Print the elements of the 2D matrix in a zig-zag (snake) pattern row-wise.

Sample Input:

3 3

1 2 3

4 5 6

7 8 9

Sample Output:

1 2 3 6 5 4 7 8 9

16. Sum of Upper and Lower Triangle

Difficulty: Medium

For a square matrix, print the sum of upper and lower triangle elements.

Sample Input:

3 3

1 2 3

4 5 6

7 8 9

Sample Output:

Upper Triangle Sum = 26

Lower Triangle Sum = 26

17. Check Symmetric Matrix

Difficulty: Medium

Check if a square matrix is symmetric (transpose is same as original).

Sample Input:

3 3

1 2 3

2 4 5

3 5 6

Sample Output:

Yes, it is symmetric.

18. Replace Zeros with Ones

Difficulty: Easy

Modify the matrix to replace all 0s with 1s.

Sample Input:

2 2

0 2

3 0

Sample Output:

1 2

3 1

19. Count Prime Numbers

Difficulty: Medium

Count how many prime numbers are present in a given 2D matrix.

Sample Input:

2 3

1 2 3

4 5 6

Sample Output:

Prime Count = 3

20. Wave Print (Column-wise Zigzag)

Difficulty: Hard

Print the elements column-wise in a wave pattern (down then up).

Sample Input:

3 3

1 2 3

4 5 6

7 8 9

Sample Output:

1 4 7 8 5 2 3 6 9